

Data cleaning with pandas

Module 2 - Lesson 2

Overview

Real data is messy. pandas provides the tools to load, inspect, and clean tabular data: missing values, wrong types, duplicates, and outliers.

Key Concepts

- DataFrames are labeled tables; read_csv loads them from files.
- df.info(), df.describe(), and df.isna().sum() reveal the shape and health of data.
- Handle missing values by dropping (dropna) or filling (fillna) based on context.
- Convert and validate types with astype and pd.to_datetime.
- Remove duplicate rows with drop_duplicates.
- Detect outliers with describe(), boxplots, and domain knowledge before deciding.

Example

```
import pandas as pd

df = pd.read_csv("sales.csv")
df["date"] = pd.to_datetime(df["date"])
df = df.drop_duplicates()
df = df.dropna(subset=["amount"])
df["amount"] = df["amount"].astype(float)
print(df.isna().sum())
```

Summary

Cleaning turns messy raw data into a reliable table. Inspect with describe/isna, then fix missing values, types, duplicates, and outliers with clear, documented decisions.

Practice Checklist

- Load a dataset and report its missing values and dtypes.
- Clean it: drop duplicates, fix dates, and handle nulls.
- Document each cleaning decision in a notebook cell.